SINGAPORE SCHOOL KINDERLAND

Computer Science — Master Study Material & Activity Guide

Cambridge IGCSE & O Level / Secondary 2 • Topic 1.3: Data Storage, Calculations & File Compression

1. File Size Calculations (Images & Audio)

Computers calculate the uncompressed size of digital images and audio files using precise formulas:

A. Bitmap Image File Size Formula:
File Size (in bits) = Image Width (pixels) × Image Height (pixels) × Colour Depth (bits per pixel)

B. Sound / Audio File Size Formula:
File Size (in bits) = Sample Rate (Hz) × Sample Resolution (bits) × Length (seconds) × Channels
(Note: Mono recording = 1 channel; Stereo recording = 2 channels)

Standard Unit Conversions (IEC Binary Standard)

Conversion Step	Mathematical Operation	Reason / Definition
Bits to Bytes	Divide by **8**	1 Byte = 8 bits
Bytes to KiB (Kibibytes)	Divide by **1,024**	1 KiB = 1,024 Bytes
Bytes to MiB (Mebibytes)	Divide by **1,024 × 1,024 = 1,048,576**	1 MiB = 1,024 KiB = 1,048,576 Bytes
Bytes to GiB (Gibibytes)	Divide by **1,024 × 1,024 × 1,024 = 1,073,741,824**	1 GiB = 1,024 MiB

Textbook Example 3 — Audio CD File Size Calculation (Page 33):
"An audio CD has a sample rate of 44,100 Hz and a sample resolution of 16 bits. The music being sampled uses two channels to allow for stereo recording. Calculate the file size for a 60-minute recording in MiB."
Step-by-Step Working:

1. **Convert time to seconds:** 60 minutes × 60 seconds = **3,600 seconds**.
2. **Calculate raw sample size:** 44,100 Hz × 16 bits × 3,600 s = **2,540,160,000 bits**.
3. **Multiply by 2 for Stereo:** 2,540,160,000 × 2 channels = **5,080,320,000 bits**.
4. **Convert bits to Bytes:** 5,080,320,000 ÷ 8 = **635,040,000 Bytes**.
5. **Convert Bytes to MiB:** 635,040,000 ÷ (1,024 × 1,024) = 635,040,000 ÷ 1,048,576 = **605.6 MiB (approx. 605 MiB)**.

2. Activity 1.16 Fully Worked Examination Solutions

Step-by-step solutions for all four examination questions from textbook Activity 1.16:

Activity 1.16 — Question 1 (Image Size in MiB):
"A camera detector has an array of 1920 by 1536 pixels. A colour depth of 16 bits is used. Calculate the size of a photograph taken by this camera, giving your answer in MiB."

- **Step 1 (Total Pixels):** 1920 × 1536 = 2,949,120 pixels.
- **Step 2 (Size in Bits):** 2,949,120 pixels × 16 bits = 47,185,920 bits.
- **Step 3 (Size in Bytes):** 47,185,920 ÷ 8 = 5,898,240 Bytes.
- **Step 4 (Size in MiB):** 5,898,240 ÷ 1,048,576 = **5.625 MiB (or 5.63 MiB)**.

Activity 1.16 — Question 2 (Photos Stored on a 16 GiB Card):
"Photographs are taken by a smartphone detector with a 1024 × 1536 pixel array and 24-bit colour depth. How many photographs could be stored on a 16 GiB memory card?"

- **Step 1 (Size of 1 Photo in Bytes):** (1024 × 1536 × 24) ÷ 8 = 1,572,864 × 3 = **4,718,592 Bytes (4.5 MiB)**.
- **Step 2 (Memory Card Capacity in Bytes):** 16 × 1,024 × 1,024 × 1,024 = **17,179,869,184 Bytes (16,384 MiB)**.
- **Step 3 (Number of Photos):** 17,179,869,184 ÷ 4,718,592 = **3,640.88 → 3,640 complete photographs**.

Activity 1.16 — Question 3 (Audio Sampling & Recording Size):
"Audio is sampled at 44.1 kHz using 8 bits and two channels. Calculate: (a) size of a 1-second sample in bits, (b) size of a 30-second audio recording in MiB."

- **(a) 1-Second Sample (in bits):** 44.1 kHz = 44,100 Hz.
  Size = 44,100 × 8 bits × 1 s × 2 channels = **705,600 bits**.
- **(b) 30-Second Recording (in MiB):**
  Total bits = 705,600 × 30 = 21,168,000 bits.
  Bytes = 21,168,000 ÷ 8 = 2,646,000 Bytes.
  MiB = 2,646,000 ÷ 1,048,576 = **2.52 MiB**.

Activity 1.16 — Question 4 (Songs Stored on a 740 MiB Music CD):
"A typical song is 3 minutes 30 seconds. Each song is sampled at 44.1 kHz, 16 bits per sample, using two channels. Calculate how many typical songs could be stored on a 740 MiB CD."

- **Step 1 (Length in Seconds):** (3 × 60) + 30 = **210 seconds**.
- **Step 2 (Size of 1 Song in Bytes):** (44,100 × 16 × 210 × 2) ÷ 8 = 296,352,000 ÷ 8 = **37,044,000 Bytes**.
- **Step 3 (Size of 1 Song in MiB):** 37,044,000 ÷ 1,048,576 = **35.328 MiB per song**.
- **Step 4 (Number of Songs on 740 MiB CD):** 740 ÷ 35.328 = **20.94 → 20 complete songs**.

3. Data Compression: Concepts & Benefits (Section 1.3.3)

Because raw audio (605 MiB/hour) and raw bitmap images take up enormous storage, files must be **compressed**.

> **Definition — Data Compression:**
> The process of reducing file size using algorithms that either eliminate redundant repeating data or discard unnoticeable detail.

Why Compress Files? (Five Core Benefits)

1. **Save Storage Space:** Frees capacity on HDDs, SSDs, smartphones, and memory cards.
2. **Enable Smooth Streaming:** Prevents buffering freezes when streaming audio or video.
3. **Faster Uploads & Downloads:** Fewer packets are transmitted, speeding up network transfers.
4. **Bandwidth Efficiency:** *Bandwidth* is the maximum rate of data transfer across a network in bits per second (bps). Compressed files use less bandwidth, allowing more simultaneous network users.
5. **Cost Savings:** Reduces cloud storage fees and avoids ISP download quota penalties.

4. Lossy vs. Lossless File Compression (Section 1.3.4)

Feature	Lossy Compression	Lossless Compression
How it Works	Permanently eliminates unnecessary data that human senses cannot perceive.	Encodes identical repeating data without losing any binary bits.
Restoration	**Cannot be reconstructed** to the original uncompressed state.	**100% fully reconstructed** bit-for-bit to original state.
Size Reduction	**Very High** (up to 85%–90% reduction).	**Moderate** (typically 30%–50% reduction).
File Formats	**MP3** (Audio), **MP4** (Video), **JPEG** (Images).	**PNG**, **GIF**, **ZIP**, **FLAC**, Run-Length Encoding.
Application	Photographs, video streaming, music tracks.	Spreadsheets, databases, program code (`.exe`), text documents.

5. Common Lossy Algorithms: MP3, MP4 & JPEG

- **MP3 (Audio):** Reduces size by ~90% using **perceptual music shaping**:
  1. Removes inaudible frequencies outside the human ear range (below 20 Hz and above 20 kHz).
  2. Eliminates quieter sounds played at the same time as louder sounds (auditory masking).
- **MP4 (Multimedia):** Compresses multimedia (audio, video, photos, animations) for internet streaming.
- **JPEG (Images):** Divides camera raw images into **8 × 8 pixel blocks**. Exploits the fact that the human eye detects brightness variations much better than colour shade variations. Separates colour from brightness and discards subtle colour details without noticeable deterioration.

6. Lossless Compression & Run-Length Encoding (RLE)

Run-Length Encoding (RLE):
A reversible, lossless compression technique that reduces adjacent repeating identical data units into two values: **[Run Count]** followed by **[Data Code]**.

Worked Example 1 — Text RLE:
String: `aaaaabbbbccddddd` (16 characters = 16 Bytes uncompressed)
• **RLE Pairs:** `05 97  04 98  02 99  05 100` (where 97='a', 98='b', 99='c', 100='d').
• **Compressed Size:** 4 pairs × 2 bytes = **8 Bytes (50% reduction)**.

Worked Example 2 — Mixed Data with Flag Byte (255):
When data does not repeat (e.g. `abcdefgh`), standard RLE would double file size. To prevent this, a **Flag Byte (255)** signals that a run follows. Single non-repeating characters are stored directly with no count byte.

Worked Example 3 — Monochrome Image Grid (Letter 'F'):
An 8 × 8 pixel grid (64 pixels = 64 bytes uncompressed) displaying the letter 'F' is encoded row-by-row into runs of Black (B) and White (W) pixels: `8B 2B 6W 2B 6W 6B 2W 2B 6W 2B 6W 8W 8W` (13 pairs = 26 bytes, saving 59% space).

7. Quick Revision Checklist & Summary

Concept	Key Takeaway
Bandwidth	Maximum rate of data transfer across a network in bits per second (bps).
Image Size Formula	Width (px) × Height (px) × Colour Depth (bits) ÷ 8 = Size in Bytes.
Sound Size Formula	Sample Rate (Hz) × Resolution (bits) × Time (s) × Channels ÷ 8 = Size in Bytes.
Lossy Target	Multimedia files (MP3, MP4, JPEG) where human senses cannot detect discarded detail.
Lossless Target	Executable files (`.exe`), spreadsheets, text, code where 100% accuracy is required.
RLE Requirement	Effective only when there are long runs of identical repeated adjacent data units.